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## Pedestrian detection method and system


### Abstract

A vehicle system for a commercial vehicle comprises a first forward-looking camera, a service braking controller; and a driver assistance controller communicating with the first forward looking camera and the service braking controller. The driver assistance controller identifies a pedestrian as a target within the field of view of the first forward looking camera, determines a distance to the pedestrian is less than or equal to a predetermined distance, transmits a signal to the service braking controller to request deceleration of the vehicle in response to the distance to the pedestrian being less than or equal to the predetermined distance. The driver assistance controller maintains the signal to the service braking controller until the pedestrian is no longer identified as a target.



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## Background/Summary

BACKGROUND

(1) The present application relates to a system, controller and method to provide precise target information when a commercial vehicle equipped with automatic emergency braking (AEB) detects forward targets.

(2) In commercial vehicle applications, different systems control service braking, lane departure warning, parking and automatic emergency braking. Different sensors, such as cameras and radars, throughout the vehicle provide the information used in these systems to determine target identification before and during implementation of automatic emergency braking.

(3) Some commercial vehicles are also equipped with pedestrian automatic emergency braking.

Pedestrian automatic emergency braking requires at least one forward facing camera. Pedestrians are various heights and may fall within a front blind spot of commercial vehicles when the camera is mounted behind the windshield. As the commercial vehicle moves closer to the target, the pedestrian may become fully immersed in the front blind spot. Therefore, there is interest in improving the functionality of pedestrian automatic emergency braking based on the sensor input and knowledge of the target position.

SUMMARY

(4) In accordance with one embodiment, a vehicle system for a commercial vehicle comprises a first forward-looking camera, a service braking controller; and a driver assistance controller communicating with the first forward looking camera and the service braking controller. The driver assistance controller identifies a pedestrian as a target within the field of view of the first forward looking camera, determines a distance to the pedestrian is less than or equal to a predetermined distance, transmits a signal to the service braking controller to request deceleration of the vehicle in response to the distance to the pedestrian being less than or equal to the predetermined distance. The driver assistance controller maintains the signal to the service braking controller until the pedestrian is no longer identified as a target.

(5) In accordance with another embodiment, a method for controlling a commercial vehicle comprises detecting a pedestrian with at least one forward camera, determining a distance to the pedestrian is less than or equal to a predetermined distance and transmitting a signal requesting deceleration of the vehicle as long as the pedestrian is detected less than the predetermined distance.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a representation of a system on a commercial vehicle according to one example of the present invention.

(2) FIG. **2** is a representation of a vehicle equipped with the system as in FIG. **1** with a sample target.

(3) FIG. **3** is a flowchart of the operation of the system as in FIG. **1**.

DETAILED DESCRIPTION

(4) Referring to FIG. **1**, a system **10** for a commercial vehicle according to one example of this invention is shown. The vehicle system **10** includes a braking controller **12**, which controls the service brakes **26** of vehicle under certain conditions. Braking controller **12** may include functionality to control anti-lock braking and stability control. Braking controller **12** responds to signals from sensors connected directly to the braking controller **12** or responds to messages received from other controllers on the vehicle. The braking controller **12** transmits signals to activate electropneumatic devices at each wheel end to assist in slowing and stopping the vehicle before the driver intervenes or in addition to driver intervention.

(5) The system **10** includes a driver assistance controller, such as automatic emergency braking (AEB) controller **14**, which transmits messages regarding potential obstacles in front of the vehicle. The AEB controller **14** may receive information about the obstacles from sensors affixed to the vehicle and looking to the front of the vehicle, such as forward-looking camera **18**, radar **20** and other sensors **22**. Specifically, the AEB controller **14** is looking for other vehicles traveling on the same roadway that may be slowing down or moving into the host vehicle's lane of travel. In general, the AEB controller **14** identifies targets based on the trajectory of the host vehicle and potential targets in the path of the host vehicle. A following distance alert is triggered when the host vehicle is within a pre-programmed distance from the target. Automatic emergency braking is requested by the AEB controller **14** when a preselected distance between the host vehicle and the

target is breached.

(6) A first camera **18** has a defined area of view of the front of the vehicle. The area of view is dependent upon the height at which the camera **18** is installed in the cab and the length of the hood of the vehicle. The camera **18** will use processing algorithms to identify whether a forward obstacle is another vehicle, a pedestrian or a traffic sign, for example. Other cameras may be installed at other locations on the vehicle, such as a driver-facing camera or a side object detection camera. The radar **20** is also positioned to scan the forward area of the vehicle for potential obstacles in the path of travel of the vehicle. The AEB controller **14** uses information from both the camera **18** and radar **20** to determine size, type and speed of targets in the path of travel of the vehicle.

(7) Pedestrian AEB is a subset of AEB that uses a forward-facing camera specifically to identify pedestrians near the vehicle. Algorithms inside the camera determine the pedestrian distance from the vehicle and transmit the information to the AEB controller **14**.

(8) The system **10** includes a parking brake controller **16**, which controls the parking brakes **28** of the vehicle so that the vehicle remains stationary. The parking brake controller **16** may receive signals regarding the driver's desire to park the vehicle as well as signals from other controllers to keep the vehicle stationary.

(9) The braking controller **12**, the AEB controller **14** and the parking brake controller **16** communicate on a vehicle serial communication bus **20**. Messages may be transmitted and received using a known protocol, such as SAE J1939, on the communication bus **30**.

(10) The system **10** includes a driver interface **24**. The driver interface may include a switch to turn on or off certain features of system **10**. The driver interface **24** may also include a lamp or display to indicate the status of the system **10**.

(11) The AEB controller **14** transmits messages on the communication bus **30** when the controller **14** determines that service braking may be necessary to mitigate a collision with a detected object. The level of braking requested depends on the velocity of the vehicle and the location of the target. The level of braking may also depend on the available braking power within the service braking system of the vehicle. For example, in a pneumatic braking system, the level of braking available is limited by the amount of pressure stored in the service braking reservoirs. The braking controller **12** responds to the messages by activating the service brakes automatically to slow or stop the vehicle. When the target is no longer being tracked, the AEB controller **14** may discontinue sending messages requesting service brake application. The parking brake controller **16** may be used to keep the vehicle stationary after a service brake application, depending on the messages from the AEB controller **14**.

(12) In another example, the functions of each controller **12**, **14**, **16** may be combined in a single controller.

(13) Therefore, a vehicle system for a commercial vehicle comprises a first forward-looking camera, a service braking controller; and a driver assistance controller communicating with the first forward looking camera and the service braking controller. The driver assistance controller identifies a pedestrian as a target within the field of view of the first forward looking camera, determines a distance to the pedestrian is less than or equal to a predetermined distance, transmits a signal to the service braking controller to request deceleration of the vehicle in response to the distance to the pedestrian being less than or equal to the predetermined distance. The driver assistance controller maintains the signal to the service braking controller until the pedestrian is no longer identified as a target.

(14) FIG. **2** depicts a vehicle **40** having the system **10**. The vehicle **40** is designed with the engine in the forward position, which creates a blind spot directly in front of the vehicle. The camera **18**, shown as forward-facing camera mounted in the cab of the vehicle, has a field of view **44** in front of the vehicle **40**, shown by the shaded area emanating from the forward-looking camera **18** located in the cab of vehicle **40**. Part of the view of the camera **18** is blocked by the hood of the vehicle **40**, creating a blind spot for the camera, not just the driver. The radar **20** also has a field of view **46**,

shown by the lines emanating from the radar **20** located on the fender of the vehicle **40**. However, a radar is typically useful in identifying metallic objects, such as vehicles and signs, rather than pedestrians.

(15) The AEB controller **14** identifies a first target as pedestrian **42** using the camera **18**. The pedestrian **42** may be a person of small stature, such as a child. For example, the height of the child may be four feet, which is lower than the height of a typical hood of a commercial vehicle.

(16) The camera **18** calculates that the pedestrian **42** is a distance d**2** from the front of the vehicle **40**. In one example, the distance d**2** is about nine feet from the front fender of the vehicle **40**. The pedestrian **42** is within the field of view **44** of the camera **18** so the entire pedestrian body is identified by the camera **18**. Dependent on the velocity of the vehicle **40**, the AEB controller **14** may transmit messages to the service brake controller **12** to slow or stop the vehicle to mitigate any collision with the pedestrian **42**.

(17) The pedestrian **42** may move to a new location, d**1**. The pedestrian **42** may have moved or the change in distance between the vehicle **40** and the pedestrian **42** may be due to the movement of the vehicle **40**. The pedestrian **42** is mostly within the field of view **44** of the camera **18**, but the entire pedestrian body is not identified by the camera **18**. Dependent on the velocity of the vehicle **40**, the AEB controller **14** will transmit messages to the service brake controller **12** to slow or stop the vehicle **40** to mitigate any collision with the pedestrian **42**.

(18) The pedestrian **42** may move to a new location, do. Without this invention, the AEB controller **14** may deactivate any service brakes since the pedestrian **42** is no longer within the field view of the camera **18** and therefore is no longer identified as a target. With this invention, as the pedestrian **42** moves in and out of the field of view **44** of the camera **18**, the AEB controller **14** maintains the service brake request as active. The AEB controller **14** uses historical data of the position of the pedestrian **42** when the pedestrian **42** is no longer within the field of view of the camera **18**. If the pedestrian **42** is not seen by camera **18** as moving outside of the blind spot, then the AEB controller **14** will request that the parking brake controller **16** activate the parking brakes to keep the vehicle **40** from moving. The AEB controller **14** assumes that the pedestrian **42** is in the same position since they were not detected as moving out of the blind spot.

(19) While the pedestrian **42** is being reliably detected by the camera **18**, the AEB controller **14** includes the logic to manage the dynamics of the event, such as host vehicle velocity, pedestrian velocity, host vehicle acceleration, longitudinal distance to determine the likelihood that under the dynamic condition the pedestrian **42** will enter the front blind spot. When this determination is made, the AEB controller **14** may set an internal flag. As the event continues and if the pedestrian detection becomes unreliable because the pedestrian **42** is moving into the front blind spot, the AEB controller **14** shall continue to transmit the braking request to the service braking controller **12**.

(20) In another embodiment, an alert may be given to the driver of the vehicle **40** via the driver interface **24** when the AEB controller **14** determines that the pedestrian **42** remains within the blind spot.

(21) In another embodiment, the other sensors **22** may include a second forward facing camera added to the fender of the vehicle **40** near the radar **20**. The second camera will be used by the AEB controller **14** to confirm whether the pedestrian **42** is still within the blind spot of the first camera **18**.

(22) FIG. **3** shows a method **60** of implementing the pedestrian watch algorithm using the system **10**.

(23) In step **62**, the method **60** begins. In step **62**, the pedestrian automated emergency braking may be active. The system **10** may be deactivated in special situations such as when the driver manually disables the system, the camera **18**, radar **20** and other sensors **22** are in an error state from a failed self-diagnosis or uncalibrated or are blocked or blind. If the driver or other system on the vehicle **40** has disabled the system **10**, either through a manual switch or communication message, then the

method **60** returns to step **62**. In another embodiment, the system **10** may be activated by geolocation, such as when the vehicle **40** is entering a known school zone. If the Pedestrian AEB is activated, the method **60** continues to step **66**.
(24) In step **66**, the longitudinal distance from the vehicle **40** to the pedestrian **42** is measured via the camera **18** and/or radar **20**. If the distance is greater than or equal to d**2**, a predetermined distance, then the method **60** returns to step **62**. The system **10** assumes that at distance d**2** or greater, the pedestrian **42** is fully visible to the driver and the camera **18** such that the regular obstacle detection will be active. If the distance is less than d**2**, the method **60** continues to step **68**.
(25) In step **68**, the relative velocity of the vehicle **40** with respect to the detected pedestrian **42** is measured. If the relative velocity is greater than or equal to a predetermined threshold, the method proceeds to step **70**. The predetermined threshold velocity may be −3.5 m/s. In step **70**, the extended Pedestrian Automatic Emergency braking is set to false. In step **72**, the Pedestrian AEB continues to function without the extended mode and the service braking may be released when no target is detected. The method **60** ends at step **74**.
(26) However, If the relative velocity is less than a predetermined threshold in step **66**, the method **60** continues to step **76**. This means that the pedestrian **42** is likely moving in such a way as to be in the front blind spot of the vehicle **40**. In step **76**, the extended pedestrian AEB is set to true. If the pedestrian **42** is no longer being reliably detected as a target in step **76**, the extended pedestrian AEB remains true and Pedestrian AEB is allowed to continue to request service braking in step **78**.
(27) If the target detection is not reliable in step **78**, then the method **60** goes directly to step **82** where the vehicle is parked via a request to the parking brake controller **16**. In step **84**, the method **60** ends.
(28) A vehicle having this system is especially useful when the vehicle is passing known areas with frequent child pedestrian crossings, such as near schools. The system may be activated only when in these locations or the system may be active during the entire operation of the vehicle.
(29) Therefore, a method for controlling a commercial vehicle comprises detecting a pedestrian with at least one forward camera, determining a distance to the pedestrian is less than or equal to a predetermined distance and transmitting a signal requesting deceleration of the vehicle as long as the pedestrian is detected less than the predetermined distance.
(30) While the present invention has been illustrated by the description of example processes and system components, and while the various processes and components have been described in detail, applicant does not intend to restrict or in any way limit the scope of the appended claims to such detail. Additional modifications will also readily appear to those skilled in the art. The invention in its broadest aspects is therefore not limited to the specific details, implementations, or illustrative examples shown and described. Accordingly, departures may be made from such details without departing from the spirit or scope of applicant's general inventive concept.

## Claims

1. A vehicle system for a commercial vehicle comprising: a first forward looking camera; a service braking controller; and a driver assistance controller communicating with the first forward looking camera and the service braking controller; wherein the driver assistance controller identifies a pedestrian as a target within the field of view of the first forward looking camera; determines a distance to the pedestrian is less than or equal to a predetermined distance; transmits a signal to the service braking controller to request deceleration of the vehicle in response to the distance to the pedestrian being less than or equal to the predetermined distance; and maintains the signal to the service braking controller as long as the driver assistance controller determines that the pedestrian remains a target in response to the pedestrian no longer being detected in a field of view of the first forward camera and not detected at a distance greater than the predetermined distance.
2. The vehicle system as in claim 1, further comprising a parking brake controller in

communication with the driver assistance controller, wherein the driver assistance controller transmits a signal to the parking brake controller to park the vehicle in response to the pedestrian remaining a target.
3. The vehicle system as in claim 2, wherein a driver of the vehicle is able to override the signal to the parking brake controller.
4. The vehicle system as in claim 1, wherein the driver assistance controller maintains the signal to the service brake controller until the pedestrian is no longer identified as a target, which occurs when the pedestrian is detected at a distance greater than the predetermined distance.
5. The system as in claim 1, wherein the driver assistance controller and the service brake controller are a single controller.
6. The system as in claim 1, further comprising a second forward looking camera mounted at a location on the vehicle separate from the first forward looking camera, wherein the pedestrian is no longer a target occurs when the second forward looking camera detects the pedestrian as a distance greater than the predetermined distance.
7. A method for controlling a commercial vehicle comprising: detecting a pedestrian with at least one forward camera; determining a distance to the pedestrian is less than or equal to a predetermined distance; transmitting a signal requesting deceleration of the vehicle as long as the pedestrian is detected less than the predetermined distance; determining that the pedestrian remains less than a predetermined distance in response to the pedestrian no longer being detected by the at least one forward camera; and continuing to transmit the signal requesting deceleration.
8. The method as in claim 7, further comprising maintaining the signal requesting deceleration of the vehicle until the pedestrian is detected at greater than the predetermined distance.
9. The method as in claim 7 further comprising: transmitting a signal to engage a parking brake of the commercial vehicle until the pedestrian has been detected at greater than the predetermined distance.
10. The method as in claim 9, further comprising: overriding the signal to engage the parking brake through an action of the driver of the commercial vehicle.
11. The method as in claim 9 further comprising: detecting the pedestrian has moved to greater than the predetermined distance; and transmitting a signal to release the parking brake of the commercial vehicle.